Defining Brightness-Shape-Moisture Soil Parameters for Southern Africa From Hyperion Hyperspectral Imagery

Wessel Bonnet, Moses Azong Cho, and Cecilia Masemola

***Abstract*— An effective methodology is needed to simulate soil spectra on a large scale. The brightness-shape-moisture (BSM) radiative transfer model (RTM) is used to simulate soil spectra for different semiarid and arid biomes within Southern Africa based on hyperspectral imagery obtained from the Hyperion satellite. Such simulation based on hyperspectral data is especially relevant in light of newer hyperspectral missions, such as Prisma providing ongoing data streams. In this particular study, Hyperion's data are cleaned using the SUREHYP procedure, segmented using the simple linear iterative clustering (SLIC) algorithm, filtered to exclude photosynthetic and senescent vegetation, and parameterized via a Hyperion band calibrated BSM model lookup table to obtain simulation parameter distributions for different biomes. This provides a means to better simulate soil spectra using each biome's obtained parameter distributions in the BSM forward model.**

***Index Terms*— Africa, Hyperion, radiative transfer, soil properties.**

I. Introduction

THE maps of soil types and their biophysical and biochemical properties are essential tools in agriculture, land use planning, environmental conservation, water management, and infrastructure planning. In general, the accurate maps of soil properties over large areas can be generated through geostatistical analysis of data from numerous representative sample points. However, obtaining soil data from many field sample points is not viable because manual soil collection and laboratory analysis is a time-consuming, destructive, laborious, and expensive exercise. Remote sensing (RS) offers a more efficient, cost-effective alternative.

RS of soil properties generally follows two methods, namely, empirical and physically based modeling approaches. In the Southern African semiarid context, this is especially important, since soil erosion, highly variable rainfall patterns, and large diurnal temperature fluctuations cause variability in soil moisture contents and structural properties, and regular measurement is needed to gauge changes. Despite this, no consolidated soil spectrum mapping has been performed for the particular study area.

In literature, soil spectrum analysis for the purposes of soil moisture-level estimation has largely been done through empirical methods [1], [2], [3], [4], [5], primarily correlating visible and near-infrared soil spectra and their first-order differences or trend-removed values against soil moisture levels. Radiative transfer methods have been less common in this field, although these vector-based models are typically more flexible than the traditional spectrum simulations [6]. By contrast, radiative transfer model (RTM) methodologies have been used extensively to model leaf and canopy characteristics and have proven to show more robustness across sites and seasons compared to empirical methods [7], [8], [9]. RTM methods are often also able to directly estimate soil moisture content without additional correlation modeling, which empirical methods cannot do [10], [11].

Soil RTMs can broadly be categorized into three categories, namely, beam tracing RTMs, multiflux radiative transfer (MFRT) models, and bidirectional reflection distribution function (BRDF) RTMs. Beam tracing models analyze surface refraction and reflectances of sun-induced radiation between air and soil media and soil spectral absorptions using Fresnel equations or Snell's law. The multilayer RTM of soil reflectance (MARMIT) model was proposed by Bablet et al. [12] is based on beam tracing, and models wet soil as dry soil surrounded by liquid water film, with light refracted and scattered between the soil and the water film. The MARMIT-2 model [13] improves on this by considering the state of mixing of soil particles within the water film and calculating a mixed soil–water reflectivity, more closely aligned to the true beam soil–water interactions.

MFRT models ignore the multiple radiance interactions between the media, only considering the bidirectional incident fluxes before and after absorption, yielding a more tractable and user-friendly RTM. Sadeghi et al. [14] laid out a model for the estimation of soil moisture based on the Kubelka–Munk model [15], [16], where surface radiance scattering and absorption characteristics are deconstructed into the individual scattering and absorption properties of dry soil with surrounding moisture and air. Ou et al. [17] put forward an equation for soil thicknesses using the Kubelka–Munk model

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Wessel Bonnet and Moses Azong Cho are with the Precision Agriculture Research Group, Advanced Agriculture and Food, CSIR, and also with the Department of Plant and Soil Sciences, University of Pretoria, Pretoria 0028, South Africa (e-mail: wbonnet@csir.co.za).

Cecilia Masemola is with the Precision Agriculture Research Group, Advanced Agriculture and Food, CSIR, Tshwane, and also with the Department of Geography, University of KwaZulu-Natal, Pietermaritzburg 3201, South Africa.

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and estimated soil organic matter content based on scattering coefficients for soil spectra of different thicknesses. These models only work at specific wavelengths, decreasing their usefulness in hyperspectral imagery analysis, and making them intractable for the modeling of continuous or high spectral resolution radiance responses.

BRDF models consider incident and reflected light distributions within a hemispherical space. The most widely used of these is the Hapke model [18], [19], [20], which separates total radiance into single-scattering and multiple-scattering radiance components. Jacquemoud et al. [21] proposed the SOILSPECT RTM for soils, based on the Hapke model using a Legendre polynomial approximation to model bidirectional scattering from the soil. Both the beam tracing and the BRDF RTMs can gauge soil moisture content if dry soil spectra are known but are unable to model dry soil.

In this study, soil spectra were modeled using the brightness-shape-moisture (BSM) RTM as first used by Verhoef et al. [22] and described by Yang et al. [23], [24], which can model full spectral signatures between 400 and 2500 nm as a beam tracing method, while also having the capability of modeling dry soil spectra before gauging soil moisture content. In previous studies utilizing the BSM model, soil spectra are not explicitly modeled, but rather approximated as a combination of specified dry and wet soil spectra [25] or modeled from existing dry soil spectra [26], [27].

In terms of RTM inversion for soil spectra retrieval, Eon et al. [28] performed soil RTM inversion using the MARMIT model to estimate soil moisture in bare dunes in the USA from UAS imagery, and Bayat et al. [26] focused on retrieving land surface properties during a drought episode from Landsat imagery using BSM within a multilayer RTM, while Prikaziuk et al. [29] inverted ASD-retrieved bare soil spectra to obtain BSM parameters for SCOPE-modeled potato fields.

However, based on research and literature sources consulted, no large-scale soil spectra inversion endeavor has been undertaken using the BSM methodology applied directly to hyperspectral spaceborne satellite imagery. Furthermore, it is posited that localized BSM soil parameter distributions will perform better than worldwide soil distributions in inverting or simulating Southern African semiarid biome soil spectra. In this study, due to the rarity of hyperspectral spaceborne data, imagery obtained from the Hyperion imaging instrument is used to find BSM parameter distributions specific to the study area. Thus, the objective of the study is to explore whether soil spectra from semiarid regions in Southern Africa can be modeled more accurately from localized BSM parameters.

II. Preprocessing

The Hyperion satellite project commissioned by Earth Observing-1 ran from 2000 to 2017 and was the first spaceborne hyperspectral imaging instrument. The sensor utilized two spectrometers capturing radiance in the SWIR, VNIR, and visible wavelengths, resulting in 242 band imagery representing wavelengths of between 356 and 2577 nm, with a 10-nm spectral resolution and a 30-m spatial resolution. Hyperion imagery is offered at three processing levels, namely, L1R, L1Gst, and L1T. L1R data contain images that are radiometrically corrected, but not terrain corrected. L1T and L1Gst imageries are georeferenced and corrected radiometrically and topographically, but because of the georeferencing, the row and column orientations no longer align to track directions.

A large component of this study involved preprocessing Hyperion images to get atmospherically corrected surface reflectances. Hyperion-specific calibrations included removal of the spectral smile (desmiling) and track-direction illumination stripes (destriping) on the L1R data and alignment of coincident VNIR and SWIR radiometric bands. Once the at-satellite radiance was corrected in this way, top-of-canopy surface reflectance could be obtained through atmospheric correction algorithms. SUREHYP, developed by Miraglio and Coops [30], was used as the consolidated procedure of algorithms for the end-to-end preprocessing of the Hyperion imagery prior to analysis. The procedure is laid out as follows.

A. Band Removal and Extraction

Out of the 242 bands in the imagery, only 200 are calibrated, namely, bands 8–57 and 77–224 [31]. Other bands had to be excluded prior to preprocessing. Second, Hyperion data appear as digital number values, from which radiances could only be obtained by dividing the numbers by the scaling factors provided in the image metadata.

B. Desmiling

For the correction of spectral smiles present within Hyperion imagery, SUREHYP utilizes San and Süzen's [32] across-track correction algorithm on the raw L1R images. The data are first transformed into the minimum noise fraction (MNF) space [33], and a second-degree polynomial is fitted to each MNF band's column means. If the second-order coefficient was over three standard deviations from the band's mean, the polynomial is subtracted from every row of the band readings as $x'_{ij} = x_{ij} - p_i$, where x_{ij} and x'_{ij} are the smiled and desmiled MNF values for column i and row j and p_i is the column i's polynomial value. The desmiled MNF data x'_{ij} are then transformed back to the original spectral radiance space.

C. Destriping

The destriping algorithm described by Pal et al. [34] available in the SUREHYP procedure was used next. This algorithm has two parts. First, global destriping was performed. For each image band k, the image column means $P_{ca,ik}$ were calculated, and the breadth of the biggest trough or ridge n_c was found via the automated find_peaks algorithm in SciPy [35] with n_c given by the median of the 20% largest breadths, rounded up to a whole number. Afterward, a second-order Savitzky–Golay filter [36] with a window length of $10n_c$ was applied on each band's column means to get a smoothed curve $P_{fit,ik}$. The globally destriped image could then be found as $x'_{ijk} = x_{ijk} + (P_{fit,ik} - P_{ca,ik})$, where x_{ijk} and x'_{ijk} are the original and updated values for a pixel in row i and column j.

Second, local image striping not spanning entire columns was removed. For a given band, a moving window of size $[3n_c, 3n_c]$ was utilized to find $I_{\text{Corr_mean},ijk}$ and $I_{\text{Corr_stdev},ijk}$, the local mean and standard deviation of windowed pixels.

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Any pixel x'_{ijk} where $|x'_{ijk} - I_{\text{Corr_mean},ijk}| > I_{\text{Corr_stdev},ijk}$ was flagged as an outlier. If 90% of a moving window column's pixels were flagged in this way, the full column was considered a local stripe, and its values x'_{ijk} were replaced by $I_{\text{Corr_mean},ijk}$.

D. Alignment of VNIR and SWIR and Georeferencing

A one-pixel along-track geometric misalignment is found between the 128th and 129th pixels in Hyperion's SWIR bands. This was corrected by shifting all the SWIR array elements after 128 up by one pixel. The SWIR spectrometer is also marginally misaligned from the VNIR spectrometer [37]. The spectrally corrected L1R data are georectified to the georeferenced L1T data using the 833-nm bands and applying a homography between the images with the corners as key points.

E. Atmospheric Correction

Hyperion imagery data are captured as at-satellite top-of-atmosphere radiance and require an atmospheric RTM to calibrate the imagery to the top-of-canopy level. SUREHYP utilizes the freely available SMARTS RTM (https://www.nrel.gov/grid/solar-resource/smarts.html), a physically based model commonly used for atmospheric correction [38], [39], [40]. The Hyperion imagery metadata provided the atmospheric and illumination input variables corresponding to the image capture. These included atmospheric vapor and ozone concentrations, solar and acquisition geometries, topographic slope and altitude data, and acquisition dates. SMARTS-applicable terrain altitudes and slope angles were calculated from publicly available Google Earth Engine data. Air water vapor levels were gauged from vapor absorption spectra through a SMARTS-generated lookup table. The impact of neighboring pixels was not considered, and the top-of-atmosphere and the top-of-canopy reflectances, ρ_{TOA} and ρ_{surf}, were calculated as

$$\rho_{\text{TOA}} = \frac{\pi L d^2}{E_{\text{sun}} \cos(\theta_z)}$$
$$\rho_{\text{surf}} = \frac{\pi (L - L_{\text{cirrus}} - L_{\text{haze}})}{\tau_{gs} \left(\tau_{\text{sg}} E_{\text{sun}} \cos(\theta_z) + E_{\text{dif}} + \hat{\rho}_{\text{terrain}} E_g\right)} \quad (1)$$

where L is the at-sensor radiance, d is the correction factor for Sun–Earth distance, E_{sun} is the out-of-atmosphere irradiance, θ_z is the solar zenith angle, L_{cirrus} is the radiance of transparent thin cirrus clouds, L_{haze} is the aerosol haze radiance, τ_{gs} is the optical path transmittance between the ground and the sensor, τ_{sg} is the optical path transmittance between the sun and the ground, E_{dif} is the diffuse irradiance, $\hat{\rho}_{\text{terrain}}$ is the reflectance of the local terrain, and E_g is the ground global irradiance, which is zero when the surface is flat. SMARTS provides τ_{sg}, τ_{gs}, E_{sun}, and $E_{\text{dif}} + \hat{\rho}_{\text{terrain}} E_g$, while θ_z is a sensor acquisition parameter. The removal of the L_{cirrus} and L_{haze} signals was the next step needed for atmospheric correction to be performed.

F. Thin Cirrus and Haze Correction

Gao et al. [41] proposed a method using the 1380-nm band to correct for the presence of thin cirrus clouds. This is the method incorporated in SUREHYP and implemented in this study. ρ_{TOA} was computed for the entire radiance pixel set, and reflectances at the 1380-nm wavelength were plotted against those at other wavelengths to get the slope factor K_{a}. Cirrus reflectances could then be corrected with $x'_{ij} = x_{ij} - (\rho_{\text{TOA},1380}/K_{a,k})$, and L_{cirrus} was subtracted as per ρ_{surf} (1).

Haze removal was performed through SUREHYP's implemented dark object subtraction as formulated by Chavez et al. [42]. The basis for this technique is that if a scene is sufficiently large, it will likely include dark pixels with reflectances (and therefore measured radiances) close to zero throughout the spectrum. Minimum radiance values are stored for all bands, and since haze is caused by atmospheric photon scattering, the relative scattering follows a wavelength power scaling law λ^c, where c is a negative value informed by atmospheric haze conditions. The aerosol radiance L_{haze} could then be computed by fitting the largest possible $A\lambda^c$ function bounded above by the stored series of minimum radiances. This aerosol radiance was subtracted from the at-sensor radiance as per (1).

G. Water Vapor Retrieval

SUREHYP's water vapor retrieval method makes use of the 940- and 1120-nm bands, where vapor absorbs light. A SMARTS lookup table was created using known parameters with vapor concentrations simulated between 0 and 12 cm. Water vapor concentrations are linked with the relative spectral absorption depths, defined as the relative amplitude between the absorption band and the mean radiance of neighboring vapor nonabsorbing bands. By default, SUREHYP uses the 1120-nm band but uses the 940-nm band as a backup if the 1120-nm band is saturated. Average water vapor levels were computed for the scene by retrieving LUT parameters yielding the closest absorption depths.

H. Terrain Correction

As per the SUREHYP procedure, elevations and the slope angles and azimuths were found using another SMARTS lookup table simulated with varying terrain altitude, slope, and azimuth values. The SMARTS outputs were then used to get each pixel's irradiance values ρ_{rough}. SUREHYP uses the modified Minnaert approach suggested by Richter et al. [43] to calculate the bidirectional reflectance distribution function for non-Lambertian rough terrains. The corrected terrain reflectance ρ_{MM} was then calculated using the solar incidence angle β, the solar zenith angle θ_S, and a wavelength λ as

$$\rho_{MM} = \rho_{\text{rough}} \left(\frac{\cos(\beta)}{\cos(\beta_T)}\right)^b \quad (2)$$

where

$$\beta_T = \begin{cases} \theta_S + 20°, & \text{if } \theta_S < 45° \\ \theta_S + 15°, & \text{if } 45° \le \theta_S < 55° \\ \theta_S + 10°, & \text{elsewhere} \end{cases}$$

and

$$b = \begin{cases} 0.5, & \text{without vegetation} \\ 0.75, & \text{for vegetation; } \lambda < 720 \text{ nm} \\ 0.33, & \text{for vegetation; } \lambda \ge 720 \text{ nm.} \end{cases}$$

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Following this correction, the Hyperion image could be utilized for further processing. This started with segmenting the processed Hyperion images into smaller zones to obtain soil spectra for images containing diverse landscapes, soil types, or soil moisture content levels.

III. Algorithms

A. Simple Linear Iterative Clustering Segmentation

SLIC is a segmentation algorithm put forward by Achanta et al. [44], [45], which implements a k-means grouping method to demarcate image segments containing pixels in the same vicinity with similar spectral characteristics. These segments are called superpixels. Using superpixels was appropriate in this study, because delineating pixel groups allowed for more refinement in spectral simulation, and allowed for a more definite discrimination between different soil types within a single image. An input parameter L informs approximately how many superpixels the full image is divided into. The full image is then split into a grid of L clusters $l \in \{1, 2, \ldots, L\}$, where G represents the grid step size. For each cluster l, a cluster center $C_l \in [R_l, y_l, x_l]$ is chosen, where R_l is the full signature of pixel l of the image, while y_l and x_l represent the row and column values of l in the image. Then, for every pixel i in a 2×2 G neighborhood of C_l, the distance from i to C_l is computed, and if C_l is the cluster center to which pixel i's spectral signature is closest, i is reassigned to cluster l. The cluster centers C_l can then be recalculated as the average spectral signature and location of pixels assigned to the cluster. This process is repeated until pixel distances fall below a specified error threshold. The obtained superpixels can then be analyzed individually to gauge soil characteristics.

B. BSM Model

The BSM model [23], [24] is an RTM used to simulate soil spectra based on various physical and structural parameters. The model characterizes soil spectra through brightness and shape transformations, as well as a calibration for moisture content. First, the model simulates dry soil spectra through a linear combination of three base soil spectra calculated from the global soil vector (GSV) library developed by Jiang and Fang [6], as shown in Fig. 1. The linear combination of these spectra is given by $R_{\text{dry}} = \sum_{i=1}^{3} f_i V_i$, where V_i are the base soil vectors, and f_i are the three coefficients calculated as

$$f = \begin{bmatrix} B \cdot \sin(\varphi) \\ B \cdot \cos(\varphi) \sin(\lambda) \\ B \cdot \cos(\varphi) \cos(\lambda) \end{bmatrix}$$

where B is a dimensionless soil brightness parameter, and φ and λ inform spectral shape. The model then uses these dry vectors together with a soil moisture content paramater SM_p to simulate the wet soil reflectance spectra R_{soil} via a physically based approach as

$$R_{\text{soil}} = R_{\text{dry}} P(k=0) + \sum_{k=1}^{\infty} R_{\text{swet}}(k) P(k) \tag{3}$$

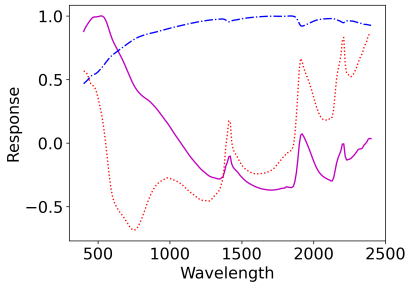


Fig. 1. GSV base soil vectors combined to model dry soils.

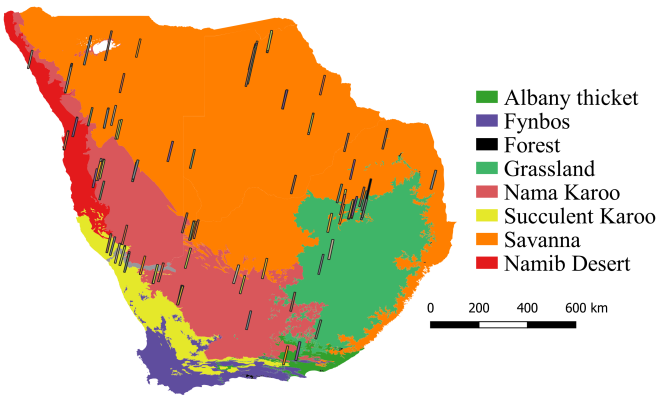


Fig. 2. Hyperion imagery within Southern African Biomes.

where

$$R_{\text{swet}}(k) = \rho_{12} + \frac{(1-\rho_{12}) \exp(-2\kappa_w k \Delta) R_{\text{bac}} (1-\rho_{21})}{1 - \rho_{21} \exp(-2\kappa_w k \Delta) R_{\text{bac}}}$$

where ρ_{12} and ρ_{21} are the air–water and water–air Fresnel reflectances, R_{bac} is the background soil reflectance, which is the dry soil reflectance spectrum plus the water–soil Fresnel reflectance, κ_w is the water absorption spectrum, Δ is the optical thickness of a single water film (usually set to a constant), and k is the number of water films around the soil particles, which follows a Poisson distribution given by

$$P(k) = \frac{e^{-\mu} \mu^k}{k!}; \quad \mu = \frac{w-5}{\text{SM}_c} \tag{4}$$

where w is the volumetric moisture carrying capacity of the soil, usually set to 25%.

IV. Simulation (Forward Modeling) of Soil Spectra

Hyperion imagery was obtained for Southern Africa from the USGS Earth Explorer database, as shown in Fig. 2. This included the countries of South Africa, Lesotho, Swaziland, Namibia, and Botswana.

Hyperion pixel responses were regularized to values between 0 and 10, and bands with anomalous spectra were excluded through simple thresholding, such that responses are between 0.01 and 0.6, and the absolute differences between consecutive bands remain below 0.1. Each image was segmented using the SLIC algorithm with 20 superpixels, as shown in Fig. 3. A total of 1765 superpixel parameter sets were obtained like this way, which were categorized into five different biomes based on the geographic location of the

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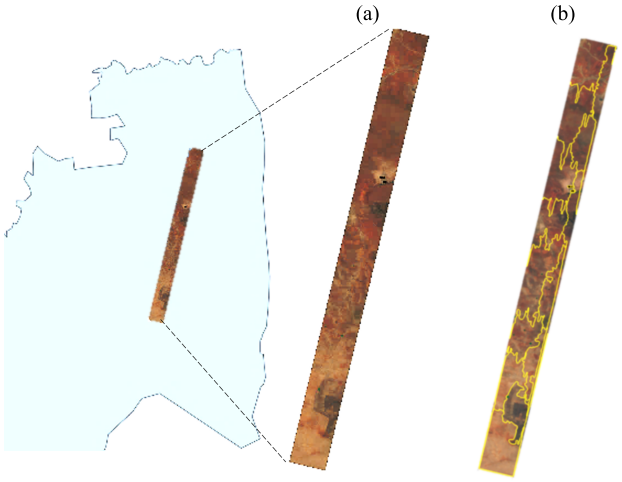


Fig. 3. (a) Preprocessed Hyperion image in Mpumalanga, South Africa. (b) SLIC-segmented Hyperion image.

TABLE I
BIOME CLASSIFICATIONS WITH TRAINING/TESTING SPLIT

Biome	Superpixels	Training set	Testing set
Desert	98	**66**	**32**
Grassland	217	**145**	**72**
Nama Karoo	446	**298**	**148**
Savanna	905	**604**	**301**
Succulent Karoo	99	**66**	**33**

original Hyperion imagery. These biomes are listed in Table I with their superpixel lookups.

Average spectra are found for each segment's pixels with cellulose absorption index (CAI) values below -1% and normalized difference vegetation index (NDVI) values between -0.046 and 0.25, where lower NDVI values represent water and higher values represent photosynthetic vegetation [47]. As in [48], Hyperion-derived CAI is given by

$$\mathrm{CAI} = \frac{\hat{\rho}_{2000} + \hat{\rho}_{2200}}{2} - \hat{\rho}_{2100} \tag{5}$$

where

$$\hat{\rho}_{2000} = \frac{\rho_{1982} + \rho_{1992} + \rho_{2002}}{3}$$
$$\hat{\rho}_{2100} = \frac{\rho_{2103} + \rho_{2113} + \rho_{2214}}{3}$$
$$\hat{\rho}_{2200} = \frac{\rho_{2194} + \rho_{2204} + \rho_{2214}}{3}.$$

The CAI measure tests for increased spectral absorption around the 2100-nm wavelength brought about by cellulose and lignin in senescent vegetation but not soil. The -1% threshold was suggested by Daughtry [49] to discriminate between their spectral signatures.

NDVI for Hyperion data is given by Pervez and Khan [46] as

$$\mathrm{NDVI} = \frac{\rho_{925} - \rho_{651}}{\rho_{925} + \rho_{651}}. \tag{6}$$

These average spectra are then smoothed using a linear Savitzky–Golay filter with a window size of 5 to decrease spectral noise. This regularizing, thresholding, and filtering process is shown in Fig. 4.

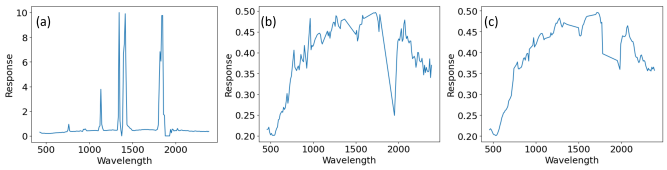


Fig. 4. Example of Hyperion superpixel spectrum. (a) Regularized. (b) Thresholded. (c) Filtered (Savitzky–Golay).

TABLE II
LOOKUP TABLE PARAMETERS FOR THE BSM MODEL

Parameter	Minimum	Maximum	Distribution
B	0.25	0.9	Uniform
φ	-30	30	Uniform
λ	80	120	Uniform
w	5%	75%	Uniform
SM_c	25%	25%	Constant
Δ	0.01cm	0.01cm	Constant

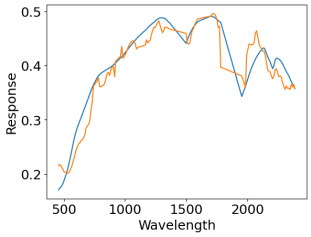


Fig. 5. Example of a processed Hyperion spectrum (orange) matched to its closest lookup table spectrum (blue).

The BSM model as implemented in the SPART package in Python [24] was used in the RTM simulation of soil spectra. The soil moisture capacity parameter SM_c was set to the default value of 25%, while the water film optical thickness parameter was set to 0.01 cm. 10 000 different soil spectra were simulated using B, φ, λ, and w ranges, as shown in Table II. The simulated lookup table spectra are then recalibrated to Hyperion spectral band responses, estimated using the provided band central wavelengths and full-width at half-maximum (FWHM) measures as provided in the Hyperion L1R metadata. The average spectra for each superpixel in the training set are then associated with parameter values from the lookup table with the closest spectra to the superpixel ones. An example of this is shown in Fig. 5, where a processed spectrum is matched to a simulated one from the lookup table to obtain its parameters. Although some spectral noise is still present, each superpixel's mean soil spectrum could be modeled to a relative root mean squared error of 1%, affirming the ability of BSM and the basis vectors to simulate Southern African spectra.

Each superpixel's mean soil spectrum could be modeled to a relative root mean squared error of 1%, affirming the ability of BSM and the basis vectors to simulate Southern African spectra. A multivariable distribution function for each biome's parameter vector $\Theta = [B, \varphi, \lambda, w]$ could then be modeled based on each biome's retrieved parameters, together with their covariances. The estimated multivariate distribution functions

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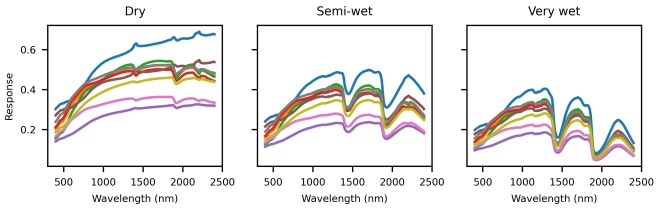


Fig. 6. Example of biome-specific simulated spectra for grassland soil in dry, semi-wet, and very wet states.

$\hat{\Theta}$ for each of the biomes are

$$\hat{\Theta}_{\text{desert}} \sim \mathcal{N}\left(\begin{bmatrix} 0.596 \\ -1.77 \\ 88.8 \\ 14.4 \end{bmatrix}, \begin{bmatrix} 0.018 & 0.253 & 0.112 & -0.855 \\ 0.253 & 10.2 & 3.62 & -42.0 \\ 0.112 & 3.62 & 15.0 & -19.3 \\ -0.855 & -42.0 & -19.3 & 192.4 \end{bmatrix}\right)$$

$$\hat{\Theta}_{\text{grassland}} \sim \mathcal{N}\left(\begin{bmatrix} 0.498 \\ -2.50 \\ 85.0 \\ 52.9 \end{bmatrix}, \begin{bmatrix} 0.012 & -0.077 & -0.112 & 0.491 \\ -0.077 & 6.38 & -0.724 & 3.21 \\ -0.112 & -0.724 & 17.6 & -33.1 \\ 0.491 & 3.21 & -33.1 & 364.4 \end{bmatrix}\right)$$

$$\hat{\Theta}_{\text{nama karoo}} \sim \mathcal{N}\left(\begin{bmatrix} 0.54 \\ -1.32 \\ 88.4 \\ 22.3 \end{bmatrix}, \begin{bmatrix} 0.012 & -0.07 & 0.142 & -0.301 \\ -0.07 & 1.93 & -1.94 & 6.33 \\ 0.142 & -1.94 & 25.7 & -48.0 \\ -0.301 & 6.33 & -48.0 & 166.9 \end{bmatrix}\right)$$

$$\hat{\Theta}_{\text{savanna}} \sim \mathcal{N}\left(\begin{bmatrix} 0.549 \\ -3.31 \\ 86.1 \\ 38.3 \end{bmatrix}, \begin{bmatrix} 0.019 & 0.06 & 0.119 & -1.23 \\ 0.06 & 19.8 & -0.379 & 7.15 \\ 0.119 & -0.379 & 25.2 & -60.0 \\ -1.23 & 7.15 & -60.0 & 50.1 \end{bmatrix}\right)$$

$$\hat{\Theta}_{\text{succ. karoo}} \sim \mathcal{N}\left(\begin{bmatrix} 0.513 \\ -1.82 \\ 90.5 \\ 14.2 \end{bmatrix}, \begin{bmatrix} 0.01 & -0.096 & -0.044 & 0.043 \\ -0.096 & 3.69 & -0.399 & 1.38 \\ 0.119 & -0.399 & 22.7 & -17.3 \\ 0.043 & 1.38 & -17.3 & 50.5 \end{bmatrix}\right) \quad (7)$$

where once again SM_c and Δ are kept constant. An example of biome-specific BSM simulated spectra is provided in Fig. 6 for grassland biome soils with soil moisture levels of 5%, 25%, and 65%, representing dry, semi-wet, and very wet soils.

As shown in Table II, a biome-specific lookup table of 1000 values was created to simulate each biome's soil spectra more accurately than a 1000-value general lookup. More specifically, for each spectrum R in the testing dataset belonging to biome b, a relative root mean relative square error (RMRSE) is found between R and $\hat{R}$ and between R and $\hat{R}_b$, where $\hat{R}$ is the closest spectrum from the full BSM table and $\hat{R}_b$ is the closest spectrum from the biome-specific lookup table. For any test spectrum j, the RMRSEs are given by

$$\text{RMRSE}_{b,j} = \sqrt{\frac{\sum_i \left(R_i - \hat{R}_{b,i}\right)^2}{n \cdot \sum_i \hat{R}_i^2}}$$

TABLE III
SPECTRA SIMULATION ERRORS UNDER FULL AND SPECIFIC PARAMETER REGIMES

Biome	RRMSE - full		RRMSE - specific		Student's
	Mean	Std error	Mean	Std error	t-test p-score
Desert	1.02%	0.03%	0.71%	0.05%	2.93×10^{-11}
Grassland	1.57%	0.12%	1.34%	0.10%	1.78×10^{-10}
Nama Karoo	1.00%	0.03%	0.84%	0.02%	9.23×10^{-45}
Savanna	2.01%	0.41%	1.29%	0.15%	8.32×10^{-3}
Succulent Karoo	0.97%	0.05%	0.8%	0.04%	2.03×10^{-11}

$$\text{RMRSE}_{f,j} = \sqrt{\frac{\sum_i \left(R_i - \hat{R}_i\right)^2}{n \cdot \sum_i \hat{R}_i^2}} \quad (8)$$

where $\text{RMRSE}_{b,j}$ and $\text{RMRSE}_{f,j}$ denote the RMRSE's under the full and biome-specific parameter regimes, respectively, and the subscript i denotes each spectral value comprising the spectrum vector.

A null hypothesis could therefore be created as follows:

$$\textbf{H0}: \exists\, b \in \text{biomes}: \mathbb{E}[\text{RMRSE}_b|b] = \mathbb{E}\left[\text{RMRSE}_f\middle|b\right]$$
$$\textbf{H1}: \forall\, b \in \text{biomes}: \mathbb{E}[\text{RMRSE}_b|b] < \mathbb{E}\left[\text{RMRSE}_f\middle|b\right]. \quad (9)$$

Since the full and biome-specific BSM simulations are used to simulate the same spectral vectors, a dependent t-test for paired samples is used to test the hypothesis, since simulation outputs are highly correlated and centered around the same target. p-scores for each biome b's t-test are provided in Table III. Assuming a 99% one-sided confidence interval is required, a p-score above 0.01 for any biome b would be required not to reject the null hypothesis. Since this is not the case, the null hypothesis is rejected, and the authors conclude that the biome-specific parameter regime does indeed yield more accurate simulations than the full parameter regime.

V. DISCUSSION AND CONCLUSION

In terms of the objective of simulating soil spectra for specified semiarid and arid biomes within Southern Africa, the BSM RTM of soils was the topic of focus. Hyperion data were preprocessed using the SUREHYP procedure, filtered via NDVI and CAI thresholding, segmented using the SLIC algorithm, and filtered and parameterized via a Hyperion band calibrated BSM model lookup table to obtain biome-specific simulation parameter distributions. These parameter distributions were tested against the full BSM parameter distributions for the simulation of actual preprocessed Hyperion spectra, and the biome-specific parameter distributions were found to yield higher simulation accuracies than the full distributions for a smaller lookup table.

This provides a strong computational advantage over full BSM soil spectral simulations and will allow for improved mapping of soil moisture content and zoning of soil types in the studied semiarid zones of Southern Africa. Since soil moisture content is one of the model's parameters, it can also be adjusted to simulate the soils at any given moisture level. The simulated pure soil spectra can then be used as a more accurate basis in canopy RTMs, such as PROSAIL and SCOPE, where local soil spectra inform the true overall

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canopy spectra separately from green or senescent plant matter for a given study area.

Although simulation results align well to Hyperion spectra, true ground truth validation for spectra, moisture, or other properties is made inviable due to the legacy nature of Hyperion data. The accuracy and efficiency of Hyperion-derived BSM parameters in modeling biome-specific soil sample spectra or tested water content remains a suitable topic for further investigation. The characteristic soil parameters may also provide further insight into biophysical and biochemical properties of soils in different biomes, and the modeling of the relationship between soil physical properties and characteristics against local BSM parameters presents a suitable topic for further research. The methodology could also be extended to retrieve BSM parameters from multispectral imagery with fewer bands.

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Wessel Bonnet received the B.Sc. degree in financial mathematics from the University of Johannesburg, Johannesburg, South Africa, in 2010, the B.Sc. degree (Hons.) in advanced mathematics of finance from the University of the Witwatersrand, Johannesburg, in 2011, and the M.Sc. degree (cum laude) in eScience from the University of the Witwatersrand, in 2011. He is currently pursuing the Ph.D. degree in plant and soil sciences with the University of Pretoria, Pretoria, South Africa.

He joined the Council for Scientific and Industrial Research, Pretoria, in 2022. His research interests include remote sensing for soil and plant biophysical and biochemical mapping, machine learning, and remote sensing image processing.



Moses Azong Cho received the B.Sc. degree in natural sciences from the University of Yaoundé, Soa, Cameroon, in 1991, the M.Sc. degree in biodiversity conservation from the University of Greenwich, London, U.K., in 2001, and the Ph.D. degree in hyperspectral remote sensing from Wageningen University, Wageningen, The Netherlands and the International Institute from Geoinformation Science and Earth Observation, Enschede, The Netherlands, in 2007.

He joined the Council for Scientific and Industrial Research, Pretoria, South Africa, in 2007. He is also an extraordinary Professor in remote sensing of vegetation with the Department of Plant and Soil Sciences, University of Pretoria, Pretoria and a Research Fellow with the University of KwaZulu-Natal, Pietermaritzburg, South Africa, since 2010. He currently serves as a Chief Research Scientist at Earth Observation Technologies, Council for Scientific and Industrial Research. His research interests include the development of remote sensing algorithms for assessing and mapping vegetation biochemical and biophysical properties and for discriminating vegetation types or species.

Dr. Cho is a member of the African Association of Remote Sensing of the Environment (AARSE) and the African Chapter President of the International Association of Landscape Ecologists (IALE). He is an Associate Editor of the *International Journal of Applied Earth Observation and Geoinformation*.



Cecilia Masemola received the B.Sc. degree in microbiology and biochemistry, the B.Sc. (Hons.) degree in environmental management, the M.Sc. degree (cum laude) in environmental science, and the Ph.D. degree from the University of South Africa, Pretoria, South Africa, in 2007, 2008, 2015, and 2019, respectively.

In 2015, she joined the Council for Scientific and Industrial Research (CSIR), initially as the Ph.D. degree student, where she became a Researcher at Earth Observation Technologies. She is currently a Senior Researcher at the CSIR's Precision Agriculture Unit. Her research interests revolve around the improvement and development of remote sensing algorithms. Specifically, she focuses on quantifying and distinguishing invasive alien plant species, such as *Acacia spp*, based on their biochemical and biophysical properties, from the leaf to the landscape level.

Dr. Masemola is an Esteemed Member of Agro-Geoinfomatics and the International Association of Landscape Ecologists (IALE).

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